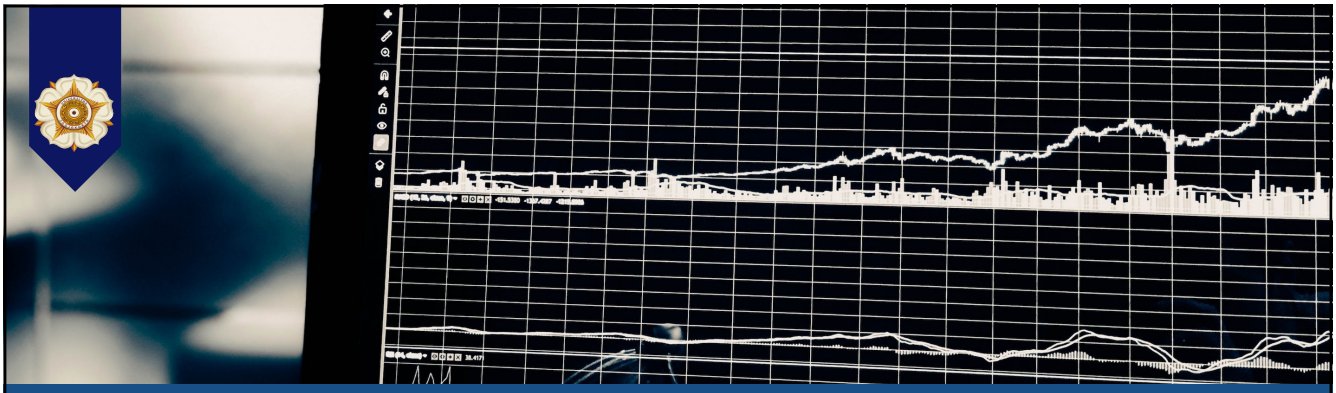


# Probability and Bayesian Reasoning

**Sunu Wibirama**
sunu@ugm.ac.id

**Department of Electrical and Information Engineering**
**Faculty of Engineering**
**Universitas Gadjah Mada**
**INDONESIA**

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1

# Why learning probability?

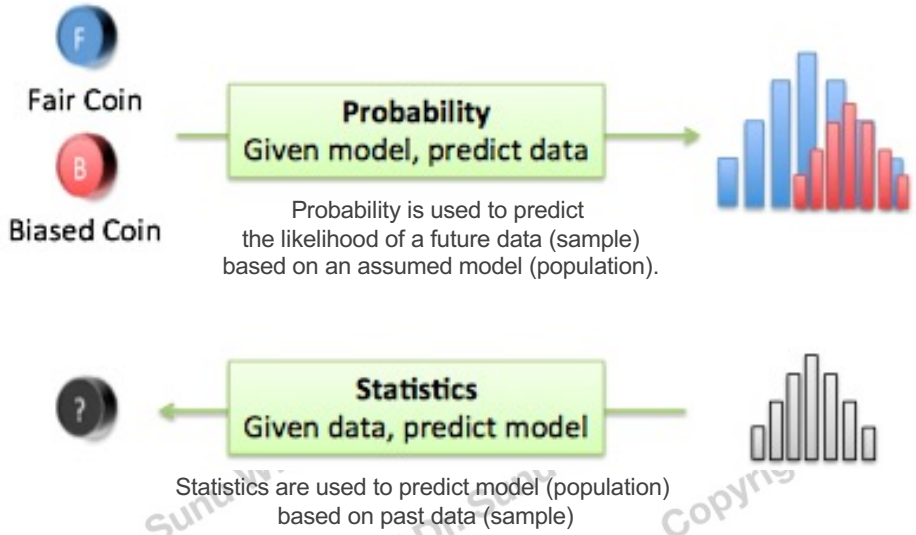
- Statistics measures observed data (sample) to describe "inner-information" (model) provided by the **already available data**
- We also interested in presentation and interpretation of **chance of outcomes (future data) with a particular assumption of the population's characteristics**
- Hence, we often deal with *experiment*: a process to generate a set of data
- Can we know exactly what will be obtained through experiment? It is difficult to get exact information of the result, but it is possible to ***predict*** the result
- That is why we have to understand basic theory of probability




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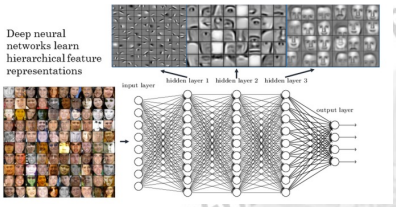
# Statistics vs probability



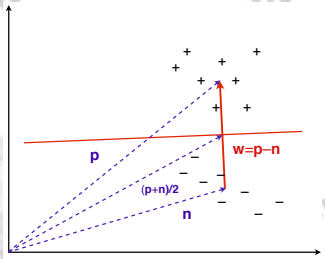

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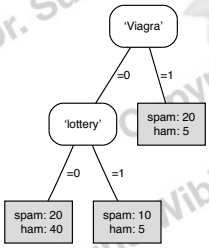
# Probability and machine learning



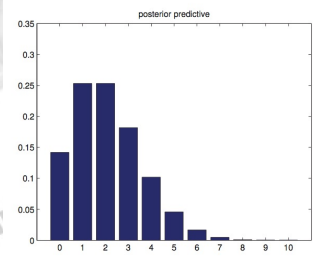

Neural network model




Geometric machine learning




Logical model/rule-based machine learning



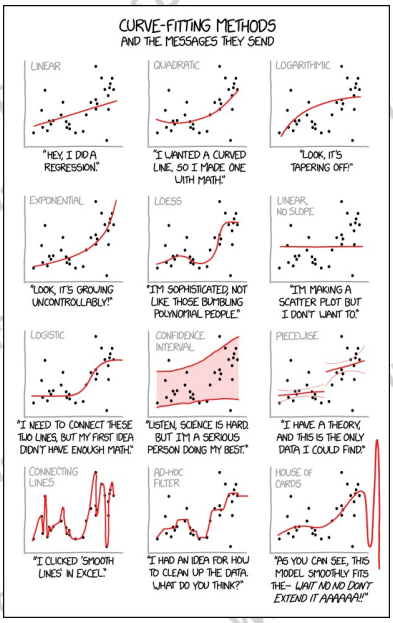

Probabilistic machine learning

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# Overview of today's lecture

- Sample size and events
- Conditional probability
- Bayesian reasoning with binary hypothesis
- Bayesian reasoning with multiple hypothesis




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Part 1

# SAMPLE SIZE AND EVENTS

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# Counting sample points

- Set of all possible outcomes of a statistical experiment is called the **sample space** (S).
- Each outcome in sample space is called: element / member / sample points
- **Example 1: Coin**
  S = { H, T }
  S : sample space when a coin is tossed once
  H : Head
  T : Tail




- **Example 2: DICE**
  $S_1$ = {1, 2, 3, 4, 5, 6}
  $S_2$ = {even, odd}



- $S_1$ provides more information than $S_2$ .
  Use sample space that provides most information of outcomes.

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# Counting sample points

- Types of sample points:
  - Finite: you can list all members
  - Infinite: you should state in *rule*
    Example: if the possible outcomes of an experiment are the set of cities in the world with a population over 1 million

$$S = \{x \mid x \text{ is a city with population over 1 million}\}$$

- Counting sample points:
  - Tree Diagram
  - Multiplication Rule
  - Permutation
  - Combination

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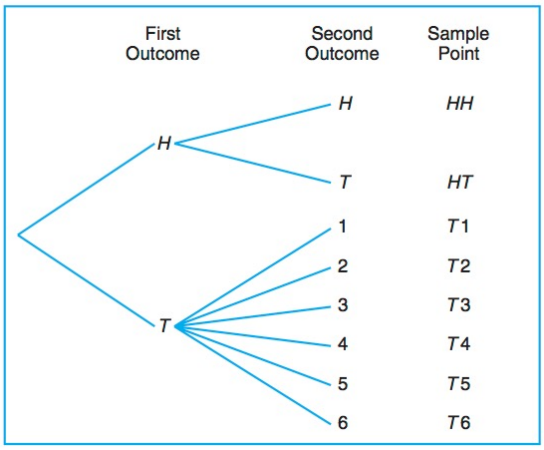

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# Tree diagram

Example:

- An experiment consists of flipping a coin and then flipping it a second time if head (H) occurs.
- If a tail (T) occurs on the first flip, then a dice is tossed once.
- To list all possible sample points, we construct tree diagram. We see that the sample space is:

S = {HH, HT, T1, T2, T3, T4, T5, T6}




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# Multiplication rule

- Should we list all sample points with tree diagram?
  No, if we have very large sample points
- We can use *multiplication rule:*
  If an operation can be performed in $n_1$ ways, and if for each of these, a second operation can be performed $n_2$ ways, and so forth, then the sequence of $k$ operations can be performed in $(n_1)(n_2)$ ……$(n_k)$ ways.
- **Example 1:**
  Compute a sample points if two dices are tossed together
- **Example 2:**
  How many menus can be formed if you have 4 soup, 3 sandwiches, 5 desserts, and 4 drinks?

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# Permutation

- Frequently, we are interested in a sample space that contains all possible orders or arrangements of a group of objects.
- We will use **permutation** to generate sample points
- A permutation is defined as an arrangement of all or part of a set of objects
- General formula of permutations of *n* distinct objects taken *r* at a time is

$$ {}_nP_r = P\begin{bmatrix} n \\ r \end{bmatrix} = \frac{n!}{(n-r)!} $$

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# Permutation

- **Example 1:**
  How many arrangements can be formed from the characters A, B, and C?

- **Answer:**
  n = 3 r = 3
  3! / (3 – 3)! = 3! / 1
  3 x 2 x 1 = 6
  ABC, BAC, CAB, ACB, CBA, BCA

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# Permutation

- **Example 2:**
  How many arrangements can be created if we choose 2 characters from four characters A, B, C, and D ?

- **Answer:**
  n = 4 r = 2
  4! / (4 – 2)! = 4 x 3 x 2! / 2! = 12
  AB, AC, AD, BA, BC, BD, CA, CB, CD, DA, DB, DC

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# Combination

- In several cases, **the order of arrangement is not so necessary to be considered**.
- We can use **combination** to compute sample points
- The number of combinations of *n* distinct objects taken *r* at a time is

$$ {}_nC_r = C\begin{bmatrix} n \\ r \end{bmatrix} = \frac{n!}{r!(n-r)!} $$

- Obviously, combination result is less than permutation's

$$ {}_nC_r \leq {}_nP_r $$

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# Combination

- **Example:**
  A young boy asks his mother to get 5 Game-Boy cartridges from his collection of 10 arcade and 5 sports games. How many ways are there that his mother can get 3 arcade and 2 sports games?

- **Answer:**
  C(10 3) = 120
  C(5 2) = 10
  120 x 10 = 1200 ways

$$ {}_nC_r = C\begin{bmatrix} n \\ r \end{bmatrix} = \frac{n!}{r!(n-r)!} $$

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# Events

- In any given experiment, we may be interested in occurrence of certain events rather than in the outcome of a specific element in the sample space.
- For instance, we are interested in the event A that the outcome when a dice is tossed is divisible by 3. Then event A is:
  A = {3, 6} of the sample space S. We can call A is subset of S
- An **event** is a subset of sample space.
- An event may be a subset that include entire sample space of S, or a no element at all.
- Event A is defined as null set $(\varnothing)$ if contains no element.
  For example: A is the event of detecting microscopic organism by the naked eye in biological experiment. Then $A = \varnothing$

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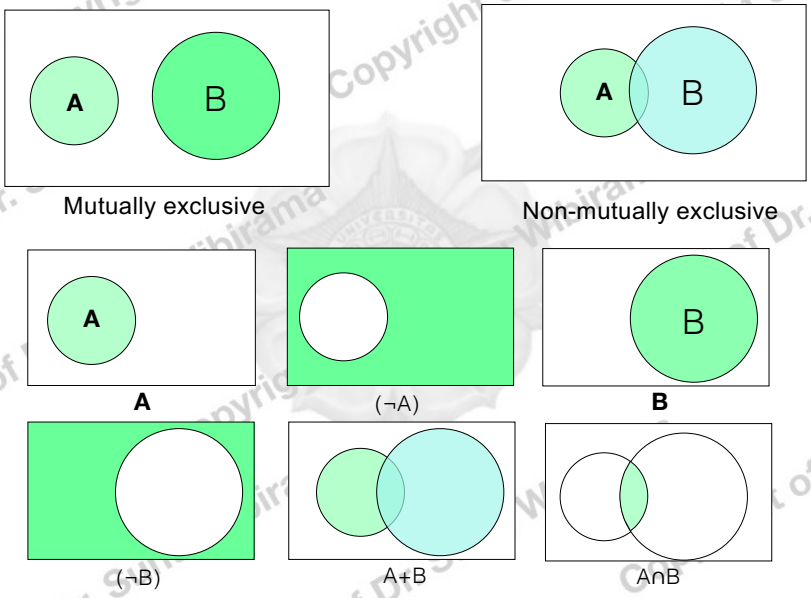
# Events

- **Field 1: complement**
  The complement of an event A with respect to S is the set of all elements of S that are not in A. We denote the complement of A by the symbol A' $(\bar{A})$
- **Field 2: intersection**
  The intersection of two events A and B, denoted by the symbol $(A \cap B)$, is the event containing all elements that are common to A and B. Sometime we can write directly as $AB$
- **Field 3: mutual exclusive**
  Two events A and B are mutually exclusive or disjoint if $A \cap B = \varnothing$ that is if A and B have no elements in common
- **Field 4: union**
  The union of the two events A and B, denoted by the symbol $A \cup B$ is the event containing all the elements that belong to **A or B or both**

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# Geometric representation of event




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Part 2

# CONDITIONAL PROBABILITY

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# Basic probability theory

- Probability can be expressed mathematically as a numerical index with a range between zero (an absolute impossibility) to unity (an absolute certainty).
- Most events have a probability index strictly between 0 and 1, which means that each event has at least two possible outcomes: **favourable outcome or success**, and **unfavourable outcome or failure**.

$$p(success) = \frac{the\ number\ of\ success}{the\ number\ of\ possible\ outcomes}$$

$$p(failure) = \frac{the\ number\ of\ failure}{the\ number\ of\ possible\ outcomes}$$

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# Basic probability theory

- If ***s*** is the number of times ***success can occur***, and ***f*** is the number of times ***failure can occur***, then

$$p(success) = p = \frac{s}{s+f}$$

$$p(failure) = q = \frac{f}{s+f}$$

$$p + q = 1$$



$$P(\text{heads}) = \frac{1}{2} = 0.5$$

- If we throw a coin, the probability of getting a head will be equ the probability of getting a tail. In a single throw, $s = f = 1$, $s + f = 2$, and therefore the probability of getting a head (or a tail) is 0.5.

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# Probability of an event

- The probability of an event A is the sum of the weights of all sample points in A. Therefore:

$$0 \le P(A) \le 1, \quad P(\varnothing) = 0, \quad \text{and} \quad P(S) = 1$$

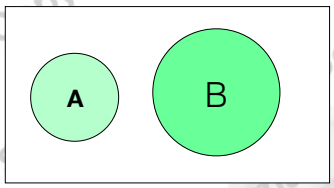
- **Example:**
  A coin is tossed twice. What is the probability that at least one head occurs?
- **Solution:**
  Compute the sample space:
  S = {HH, HT, TH, TT}
  Define A = {HH, HT, TH}

$$P(A) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$$

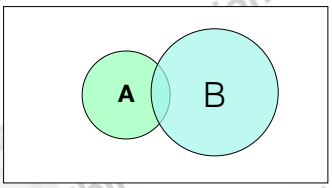
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# Geometric representation of event




Mutually exclusive




Non-mutually exclusive

Given two events, *A* and *B*, we define the probability of *A* or *B* as follows:

$$p(A \cup B) = p(A) + p(B) - p(A \cap B)$$

if A and B are non-mutually exclusive

or

$$p(A \cup B) = p(A) + p(B)$$

if A and B are mutually exclusive

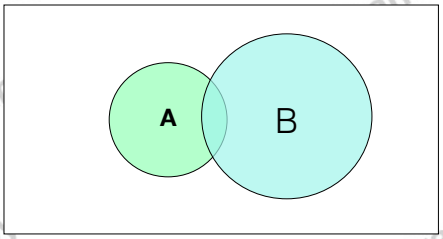
*Non-mutually exclusive:* ada peluang *event* A dan B muncul bersamaan.
Contoh: peluang munculnya *tail* dan angka 6 ketika kita melempar dadu, lalu melempar koin

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# Conditional probability

- Let *A* be an event in the world and *B* be another event. Suppose that events *A* and *B* are **not mutually exclusive / not separated**, but occur **conditionally** on the occurrence of the other.
- The probability that event *A* will occur if event *B* occurs is called the **conditional probability**.
- Conditional probability is denoted mathematically as *p*(*A*|*B*) in which the vertical bar represents "given" and the complete probability expression is interpreted as: conditional probability of event A occurring given that event B has occurred.




$$p(A|B) = \frac{\textit{the number of times A and B can occur}}{\textit{the number of times B can occur}}$$

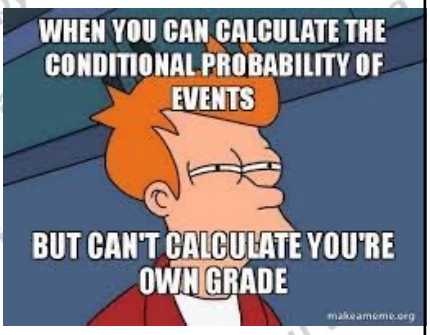
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# Conditional probability

- The number of times *A* and *B* can occur, or the probability that **both *A* and *B* will occur**, is called the **joint probability of *A* and *B***. It is represented mathematically as $p(A \cap B)$ .
- The number of ways *B* can occur is the probability of *B*: $p(B)$
- The probability of an event *A*, given that an event *B* has occurred, is called the **conditional probability of *A* given *B*** and denoted by the symbol $p(A|B)$
- $$p(A|B) = \frac{p(A \cap B)}{p(B)}$$
- The probability of an event *B*, given that an event *A* has occurred, is called the **conditional probability of *B* given *A*** and denoted by the symbol $p(B|A)$

$$p(B|A) = \frac{p(B \cap A)}{p(A)}$$




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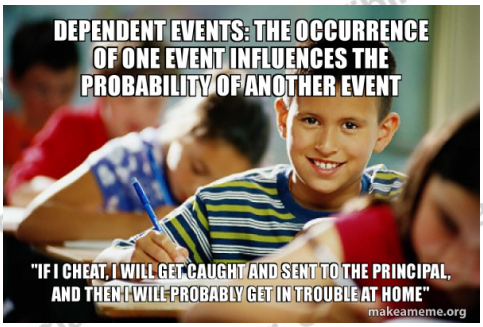
# Conditional probability and independence

Two events A and B are **independent** if and only if

$$p(B|A) = p(B)$$

$$p(A|B) = p(A)$$

Otherwise, A and B are **dependent.**




---

*Independent:*
*event* A dan B bersifat *non-mutually exclusive* (ada peluang dua-duanya muncul bersamaan), tapi kemunculan A tidak mempengaruhi B, dan juga sebaliknya. Misal: kemunculan mata dadu 6 dan bagian *tail* pada koin jika kita melempar dadu, lalu melempar koin

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# Conditional probability

Hence $p(B \cap A) = p(B|A) \times p(A)$

and $p(A \cap B) = p(A|B) \times p(B)$

Substituting the first equation into this equation

$$p(A|B) = \frac{p(A \cap B)}{p(B)}$$

$$p(A|B) = \frac{p(A \cap B)}{p(B)}$$

$$p(B|A) = \frac{p(B \cap A)}{p(A)}$$

yields the **Bayesian rule** (developed by a statistician Thomas Bayes)

$$p(A|B) = \frac{p(B|A) \times p(A)}{p(B)}$$

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Part 3

# BAYESIAN REASONING WITH BINARY HYPOTHESIS

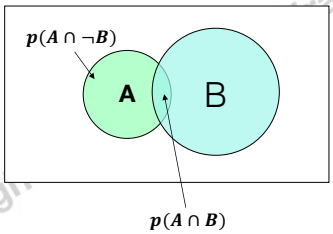
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# Bayesian reasoning

- If the occurrence of event A depends on **only two mutually exclusive events (*B* and NOT *B*)** we obtain:

$$p(A) = \underbrace{p(A|B)\times p(B)}_{p(A\cap B)} + \underbrace{p(A|\neg B)\times p(\neg B)}_{p(A\cap \neg B)}$$




where ¬ is the logical function NOT. Similarly

$$p(B) = p(B|A)\times p(A) + p(B|\neg A)\times p(\neg A)$$

- Substituting this equation into the Bayesian rule $p(A|B) = \frac{p(B|A)\times p(A)}{p(B)}$

yields: $$p(A|B) = \frac{p(B|A)\times p(A)}{p(B|A)\times p(A) + p(B|\neg A)\times p(\neg A)}$$

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# Bayesian reasoning

The Bayesian rule expressed in terms of hypothesis (***H***) and evidence (***E***) looks like this:

$$\underbrace{p(H|E)}_{\text{Posterior probability}} = \frac{\overbrace{p(E|H)}^{\text{Likelihood}}\times\overbrace{p(H)}^{\text{Prior probability}}}{\underbrace{p(E|H)\times p(H) + p(E|\neg H)\times p(\neg \mathrm{H})}_{\text{Marginal probability}}}$$



Thomas Bayes (1701-1762)

**where:**

$p(H|E)$ probability that hypothesis *H* is true given evidence *E*

$p(H)$ is the **prior probability** of hypothesis ***H* being true**;

$p(E|H)$ is the probability that hypothesis *H* being true **will result in evidence *E***;

$p(\neg H)$ is the prior probability of hypothesis *H* being false;

$p(E|\neg H)$ is the probability of finding evidence *E* even when hypothesis *H* is false.

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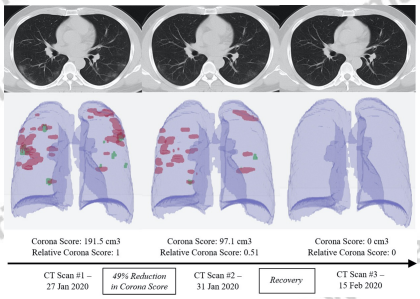
# Computer aided diagnosis for Covid-19 detection

- Recently you decide to have a CT-scan test for Covid-19. If the test is positive, what is the probability you are infected?
- Suppose you are told the test has a **sensitivity** of 80%, which means, if you are infected by Covid-19, the test will be positive with probability 0.8. In other words

$$p(x = 1|y = 1) = 0.8$$

Note:
*x* is evidence (*E*)
*y* is hypothesis (*H*)

where *x = 1* is the event the test is positive, and *y = 1* is the event you are infected by Covid-19.




Source: https://spectrum.ieee.org/hospitals-deploy-ai-tools-detect-covid19-chest-scans

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# Computer aided diagnosis for Covid-19 detection

- **Many people conclude they are therefore 80% likely to be infected by Covid-19.** But this is false! It ignores the prior probability of having Covid-19, which fortunately is quite low:

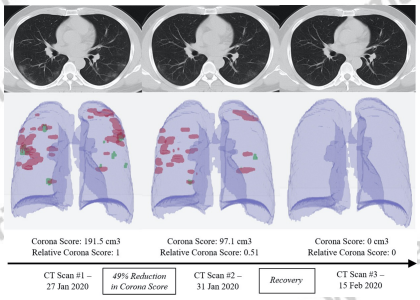
$$p(y = 1) = 0.004$$

Prior probability that someone is infected by Covid-19

- We also need to take into account the fact that the test may be a **false positive** or **false alarm**. Unfortunately, such false positives are quite likely (with current screening technology):

$$p(x = 1|y = 0) = 0.1$$

Probability that the test result is positive, although you aren't infected




Source: https://spectrum.ieee.org/hospitals-deploy-ai-tools-detect-covid19-chest-scans

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# Computer aided diagnosis for Covid-19 detection

$$p(H|E) = \frac{p(E|H) \times p(H)}{p(E|H) \times p(H) + p(E|\neg H) \times p(\neg H)}$$

Note:
*x* is evidence (*E*)
*y* is hypothesis (*H*)

- Combining these three terms using Bayes rule, we can compute the correct answer as follows:

***Posterior knowledge*** ***Likelihood*** ***Prior knowledge***

$$p(y=1|x=1) = \frac{p(x=1|y=1)p(y=1)}{p(x=1|y=1)p(y=1) + p(x=1|y=0)p(y=0)}$$

$$= \frac{0.8 \times 0.004}{0.8 \times 0.004 + 0.1 \times 0.996} = 0.031$$

where $p(y=0) = 1 - p(y=1) = 0.996$.

*Glossary:*

| | |
|---|---|
| *p*(x=1 \| y=1) | : peluang seseorang terinfeksi Covid-19 dan hasil tes positif |
| *p*(y=1) | : peluang seseorang terinfeksi Covid-19 |
| *p*(y=0) | : peluang seseorang TIDAK terinfeksi Covid-19 |
| *p*(x=1 \| y=0) | : *false alarm*, seseorang tidak terinfeksi, tapi hasil tes menyatakan ia positif terinfeksi Covid-19 |

In other words, if the test's result is positive, you only have about a 3% (0.03) chance of actually infected by Covid-19!

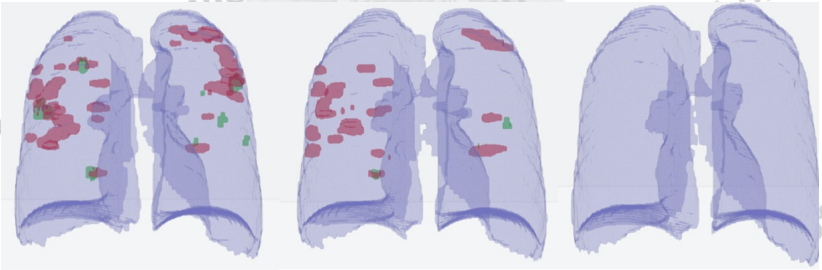
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# Computer aided diagnosis for Covid-19 detection

- The Computer Aided Diagnosis (CAD) system can then determine whether you are infected by Covid-19 using a threshold (e.g. **corona score**) on the probability value.
- For example, using corona score of 0.5 :
  - IF $p(y=1|x=1) > 0.5$

    THEN the CAD will notify the medical doctor that you are infected by Covid-19.



Corona Score: 191.5 cm3
Relative Corona Score: 1

Corona Score: 97.1 cm3
Relative Corona Score: 0.51

Corona Score: 0 cm3
Relative Corona Score: 0

Source: https://spectrum.ieee.org/hospitals-deploy-ai-tools-detect-covid19-chest-scans

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Part 4

# BAYESIAN REASONING WITH MULTIPLE HYPOTHESIS

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## Bayesian reasoning with multiple hypothesis and multiple evidences

- We can take into account both multiple hypotheses $H_1$, $H_2$,..., $H_m$ and multiple evidences $E_1$, $E_2$,..., $E_n$. The hypotheses as well as the evidences must be mutually exclusive and exhaustive/comprehensive
- Single evidence $E$ and multiple hypotheses follow:

$$p(H_i|E) = \frac{p(E|H_i) \times p(H_i)}{\sum_{k=1}^{m} p(E|H_k) \times p(H_k)}$$

- Multiple evidences and multiple hypotheses follow:

$$p(H_i|E_1\ E_2 \ldots E_n) = \frac{p(E_1\ E_2 \ldots E_n|H_i) \times p(H_i)}{\sum_{k=1}^{m} p(E_1\ E_2 \ldots E_n|H_k) \times p(H_k)}$$

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# Bayesian reasoning with multiple hypothesis and multiple evidences

- This requires obtaining the conditional probabilities of all possible combinations of evidences for all hypotheses.
- Therefore, **conditional independence** among different evidences assumed.
- Thus, instead of the unworkable equation, we attain:

$$p(H_i|E_1\,E_2\,\ldots E_n)=\frac{p(E_1|H_i)\times p(E_2|H_i)\times\ldots\times p(E_n|H_i)\times p(H_i)}{\sum_{k=1}^{m} p(E_1|H_k)\times p(E_2|H_k)\times\ldots\times p(E_n|H_k)\times p(H_k)}$$

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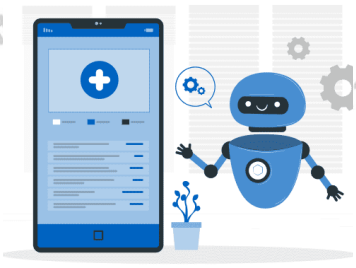

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# Ranking potentially true hypotheses

- Let us consider a simple example:
  - Suppose a medical AI system is given three conditionally independent evidences $E_1$, $E_2$,..., $E_n$, creates three mutually exclusive and exhaustive (comprehensive) hypotheses $H_1$, $H_2$,..., $H_m$, and provides prior probabilities for these hypotheses : $p(H_1)$, $p(H_2)$ and $p(H_3)$, respectively.

$H_1$= dengue $H_2$=flu $H_3$=malaria ⟶ diseases

$E_1$=headache $E_2$=cough $E_3$=fever ⟶ symptoms



  - The medical AI system also determines the conditional probabilities of observing each evidence for all possible hypotheses.

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# The prior and conditional probabilities

- $H_1$=dengue
- $H_2$=flu
- $H_3$=malaria

| Probability | Hypothesis | | |
|---|---|---|---|
| | $i=1$ | $i=2$ | $i=3$ |
| $p(H_i)$ | 0.40 | 0.35 | 0.25 |
| $p(E_1\|H_i)$ | 0.3 | 0.8 | 0.5 |
| $p(E_2\|H_i)$ | 0.9 | 0.0 | 0.7 |
| $p(E_3\|H_i)$ | 0.6 | 0.7 | 0.9 |

Peluang terjadinya kasus penyakit dengue, flu, malaria

Peluang terjadinya sakit kepala ($E_1$) pada pasien berpenyakit dengue, flu, malaria

Peluang terjadinya batuk ($E_2$) pada pasien berpenyakit dengue, flu, malaria

Peluang terjadinya demam ($E_3$) pada pasien berpenyakit dengue, flu, malaria

- $E_1$=headache, $E_2$=cough, $E_3$=fever.

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Assume that we first observe evidence $E_3$(fever) that is found in a patient.
The AI system computes the posterior probabilities of each hypothesis as:

$$p(H_i|E_3)=\frac{p(E_3|H_i)\times p(H_i)}{\sum_{k=1}^{3} p(E_3|H_k)\times p(H_k)},\quad i=1,2,3$$

thus

$$p(H_1|E_3)=\frac{0.6\cdot 0.40}{0.6\cdot 0.40+0.7\cdot 0.35+0.9\cdot 0.25}=0.34$$

$$p(H_2|E_3)=\frac{0.7\cdot 0.35}{0.6\cdot 0.40+0.7\cdot 0.35+0.9\cdot 0.25}=0.34$$

$$p(H_3|E_3)=\frac{0.9\cdot 0.25}{0.6\cdot 0.40+0.7\cdot 0.35+0.9\cdot 0.25}=0.32$$

- $H_1$=dengue
- $H_2$=flu
- $H_3$=malaria

| Probability | Hypothesis | | |
|---|---|---|---|
| | $i=1$ | $i=2$ | $i=3$ |
| $p(H_i)$ | 0.40 | 0.35 | 0.25 |
| $p(E_1\|H_i)$ | 0.3 | 0.8 | 0.5 |
| $p(E_2\|H_i)$ | 0.9 | 0.0 | 0.7 |
| $p(E_3\|H_i)$ | 0.6 | 0.7 | 0.9 |

- $E_1$=headache, $E_2$=cough, $E_3$=fever.

After evidence $E_3$ is observed, belief in hypothesis $H_2$ decreases and becomes equal to belief in hypothesis $H_1$ ($p(H_2)$ : from 0.35 to 0.34). Belief in hypothesis $H_3$ increases and even nearly reaches beliefs in hypotheses $H_1$ and $H_2$ ($p(H_3)$: from 0.25 to 0.32)

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Suppose now that we also observe evidence $E_1$ (headache) in a patient. The posterior probabilities are calculated as follows:

$$p(H_i|E_1E_3)=\frac{p(E_1|H_i)\times p(E_3|H_i)\times p(H_i)}{\sum_{k=1}^{3} p(E_1|H_k)\times p(E_3|H_k)\times p(H_k)}, \qquad i=1,2,3$$

hence

$$p(H_1|E_1E_3)=\frac{0.3\cdot 0.6\cdot 0.40}{0.3\cdot 0.6\cdot 0.40+0.8\cdot 0.7\cdot 0.35+0.5\cdot 0.9\cdot 0.25}=0.19$$

$$p(H_2|E_1E_3)=\frac{0.8\cdot 0.7\cdot 0.35}{0.3\cdot 0.6\cdot 0.40+0.8\cdot 0.7\cdot 0.35+0.5\cdot 0.9\cdot 0.25}=0.52$$

$$p(H_3|E_1E_3)=\frac{0.5\cdot 0.9\cdot 0.25}{0.3\cdot 0.6\cdot 0.40+0.8\cdot 0.7\cdot 0.35+0.5\cdot 0.9\cdot 0.25}=0.29$$

- $H_1$=dengue
- $H_2$=flu
- $H_3$=malaria

| Probability | Hypothesis | | |
|---|---|---|---|
| | $i=1$ | $i=2$ | $i=3$ |
| $p(H_i)$ | 0.40 | 0.35 | 0.25 |
| $p(E_1\|H_i)$ | 0.3 | 0.8 | 0.5 |
| $p(E_2\|H_i)$ | 0.9 | 0.0 | 0.7 |
| $p(E_3\|H_i)$ | 0.6 | 0.7 | 0.9 |

- $E_1$=headache, $E_2$=cough, $E_3$=fever.

Hypothesis $H_2$ has now become the most likely one (0.52 compared with 0.19 and 0.29).

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After observing evidence $E_2$, the final posterior probabilities for all hypotheses are calculated:

$$p(H_i|E_1E_2E_3)=\frac{p(E_1|H_i)\times p(E_2|H_i)\times p(E_3|H_i)\times p(H_i)}{\sum_{k=1}^{3} p(E_1|H_k)\times p(E_2|H_k)\times p(E_3|H_k)\times p(H_k)}, \qquad i=1,2,3$$

hence

$$p(H_1|E_1E_2E_3)=\frac{0.3\cdot 0.9\cdot 0.6\cdot 0.40}{0.3\cdot 0.9\cdot 0.6\cdot 0.40+0.8\cdot 0.0\cdot 0.7\cdot 0.35+0.5\cdot 0.7\cdot 0.9\cdot 0.25}=0.45$$

$$p(H_2|E_1E_2E_3)=\frac{0.8\cdot 0.0\cdot 0.7\cdot 0.35}{0.3\cdot 0.9\cdot 0.6\cdot 0.40+0.8\cdot 0.0\cdot 0.7\cdot 0.35+0.5\cdot 0.7\cdot 0.9\cdot 0.25}=0$$

$$p(H_3|E_1E_2E_3)=\frac{0.5\cdot 0.7\cdot 0.9\cdot 0.25}{0.3\cdot 0.9\cdot 0.6\cdot 0.40+0.8\cdot 0.0\cdot 0.7\cdot 0.35+0.5\cdot 0.7\cdot 0.9\cdot 0.25}=0.55$$

- $H_1$=dengue
- $H_2$=flu
- $H_3$=malaria

| Probability | Hypothesis | | |
|---|---|---|---|
| | $i=1$ | $i=2$ | $i=3$ |
| $p(H_i)$ | 0.40 | 0.35 | 0.25 |
| $p(E_1\|H_i)$ | 0.3 | 0.8 | 0.5 |
| $p(E_2\|H_i)$ | 0.9 | 0.0 | 0.7 |
| $p(E_3\|H_i)$ | 0.6 | 0.7 | 0.9 |

- $E_1$=headache, $E_2$=cough, $E_3$=fever.

Although the initial ranking was $H_1$, $H_2$ and $H_3$, only hypotheses $H_1$ and $H_3$ remain under consideration after all evidences ($E_1$, $E_2$ and $E_3$) were observed. Finally, the doctor should than decide whether the patient has got dengue or malaria (malaria is likely as the probability is higher than dengue, $H_3 > H_1$).

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## What about implementation on sentiment analysis using social media?

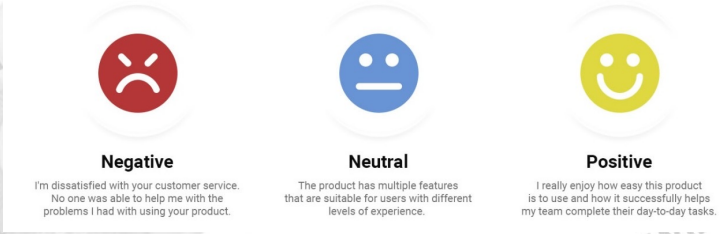



- You can do similar computation, given that :

  $H_1$= neutral
  $H_2$= positive sentiment
  $H_3$= negative sentiment

  $E_1$= occurence of word “great”
  $E_2$= occurence of word “good”
  $E_3$= occurence of word “product”
  $E_k$= occurence of word .....

The engineer should provide information of likelihood and prior probability




| Probability | Hypothesis | | |
|---|---|---|---|
| | $i=1$ | $i=2$ | $i=3$ |
| $p(H_i)$ | 0.40 | 0.35 | 0.25 |
| $p(E_1\|H_i)$ | 0.3 | 0.8 | 0.5 |
| $p(E_2\|H_i)$ | 0.9 | 0.0 | 0.7 |
| $p(E_3\|H_i)$ | 0.6 | 0.7 | 0.9 |

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## Take home messages

- Statistics and probability are commonly used in data analysis. Statistics is used for inference (determining model from data). Probability is used to predict future outcome from an assumed model (predicting data from model).
- Sample size and events are two basic concept in probability theory. Based on types of event, we can define appropriate probabilistic operation.
- Bayesian reasoning is a useful concept to predict data from evidences, given that the model follows Bayesian rules.
- Bayesian reasoning forms fundamental concept in probabilistic machine learning. Hence, an important concept in the era of Big Data and AI.

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@sunu_wibirama sunu@ugm.ac.id


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